Rho Sweep - Reference Document

Timeline in: Regressive Property Taxation with Fairness
Code in: Github repo

Overview

1. Rho selection

```
python quick_test_models.py \
  --models linear,lgbm,cov \
  --rho 3 \
  --lgbm-hyperparameter-file best_lgbm_baseline_configs.yaml \
  --lgbm-config-key test_best_r2 \
  --out-dir output/quick_test_best_lgbm_cov \
  --lgbm-n-jobs 380


python quick_test_models.py \
  --rho 3 \
  --lgbm-hyperparameter-file best_lgbm_baseline_configs.yaml \
  --lgbm-config-key test_best_r2 \
  --lgbm-n-jobs 1 \
  --models linear,lgbm,cov \
  --out-dir output/quick_test_example \
  --data-path data/CCAO/2025/training_data.parquet \
  --parallel-models \
  --sample-frac 1.0


# not used for the experiments:  --early-stopping-rounds 10 \


python quick_test_models.py \
  --rho-range "1e-1,20" \
  --rho-count 30 \
  --rho-scale log \
  --lgbm-hyperparameter-file best_lgbm_baseline_configs.yaml \
  --lgbm-config-key test_best_r2 \
  --lgbm-n-jobs 1 \
  --lgbm-n-estimators 100 \
```

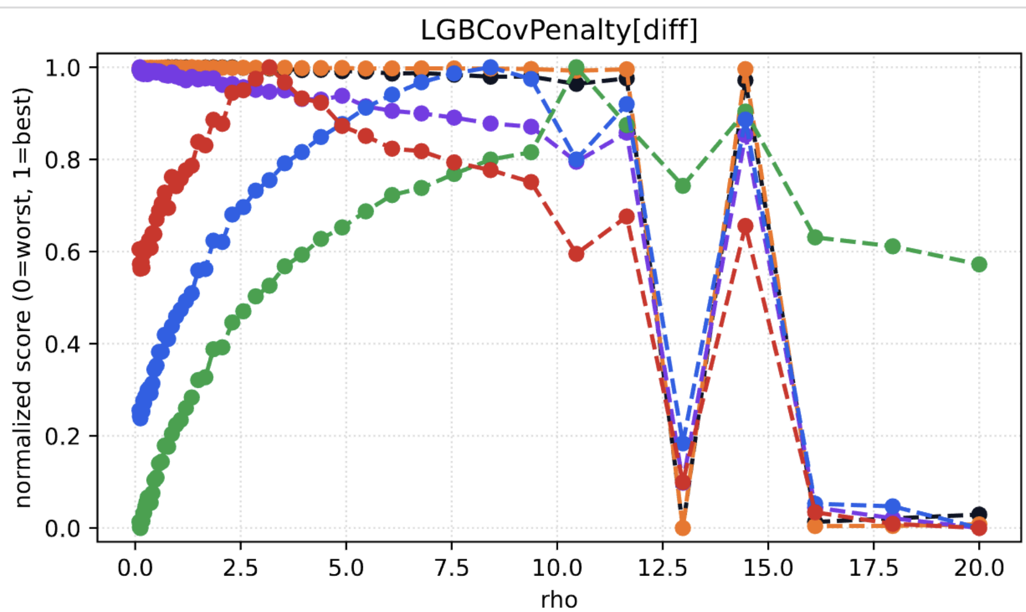
```
    --models linear,lgbm,cov \
    --out-dir output/quick_test_example \
    --data-path data/CCAO/2025/training_data.parquet \
    --parallel-models \
    --sample-frac 0.1 \
    --seed 4050 \
    --n-bootstrap-validation 5 \
    --bootstrap-block-freq M \
    --scatter-plot-max-samples 1000
```

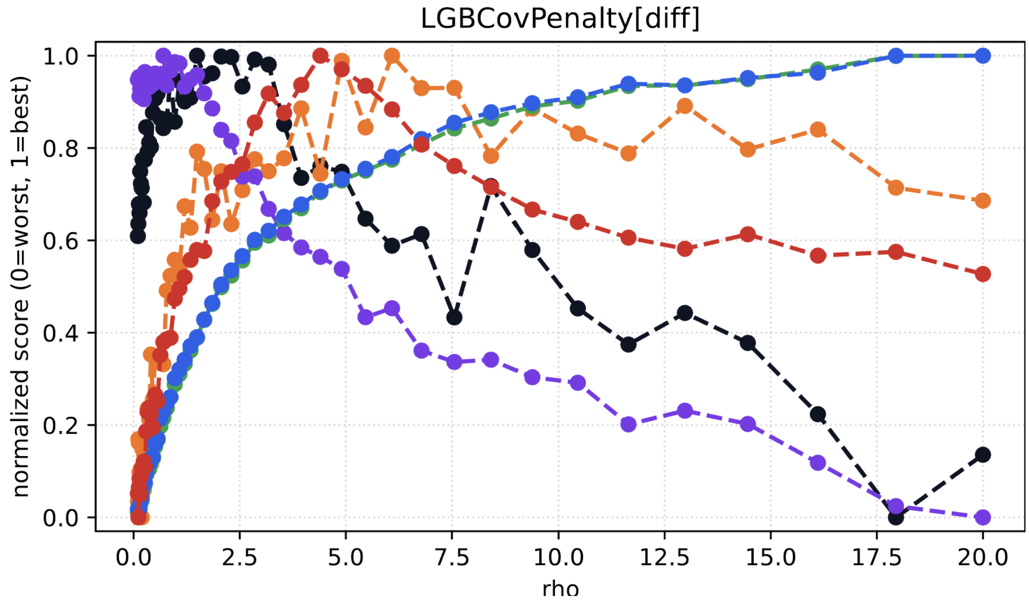
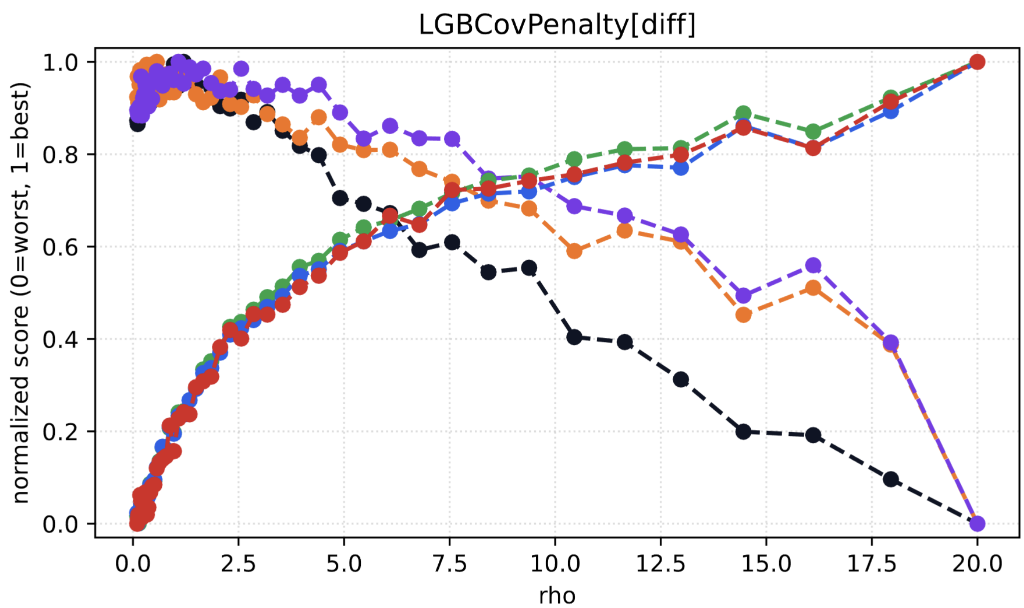
Results

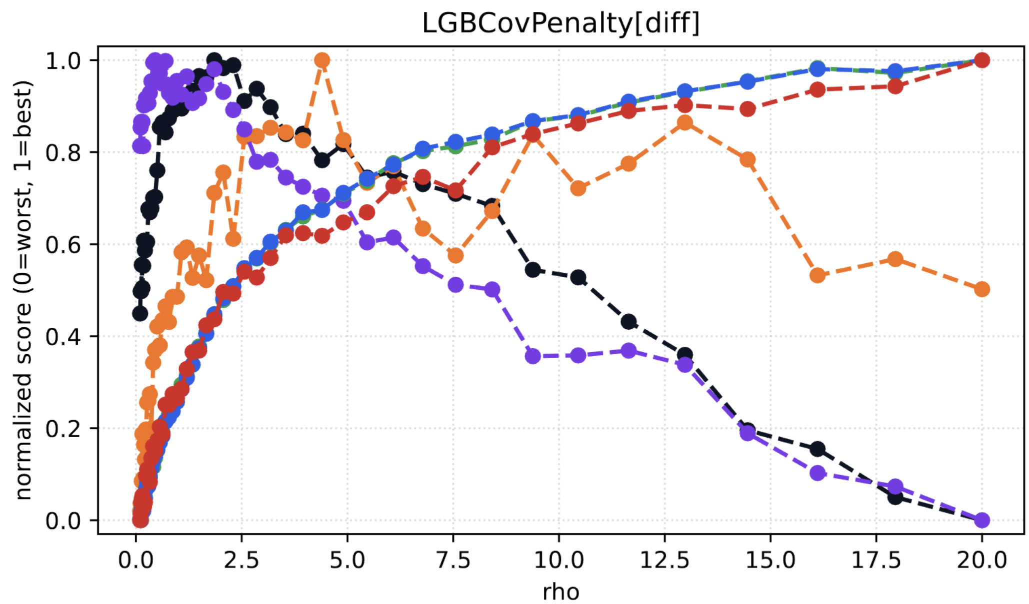
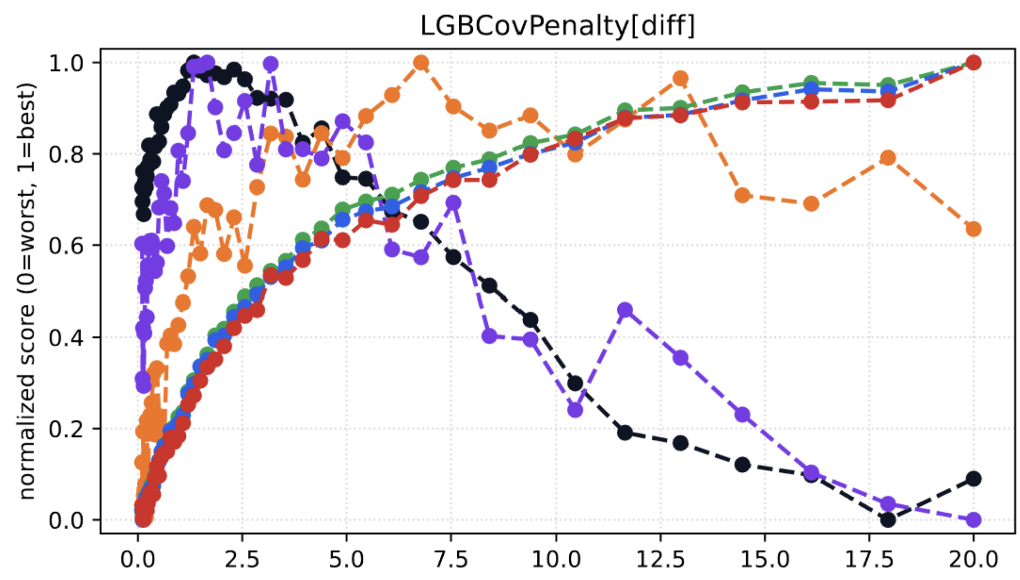
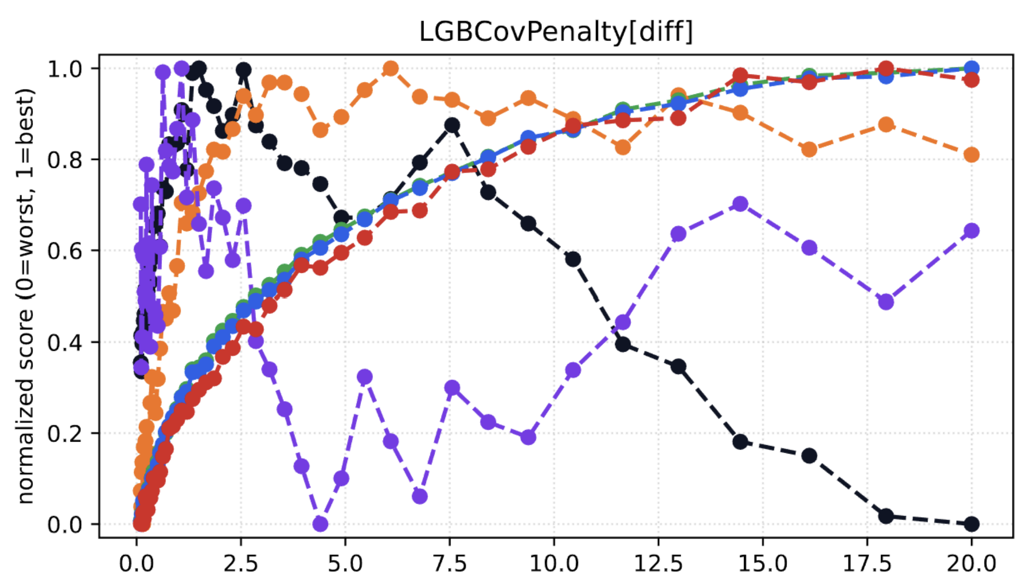
1. All Metrics Evolution w.r.t. ρ

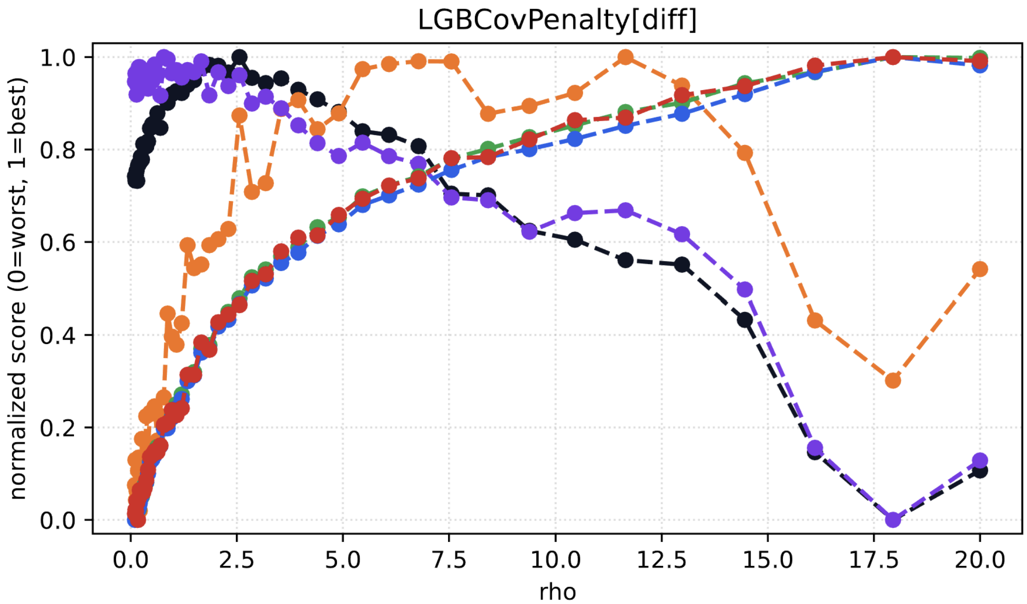
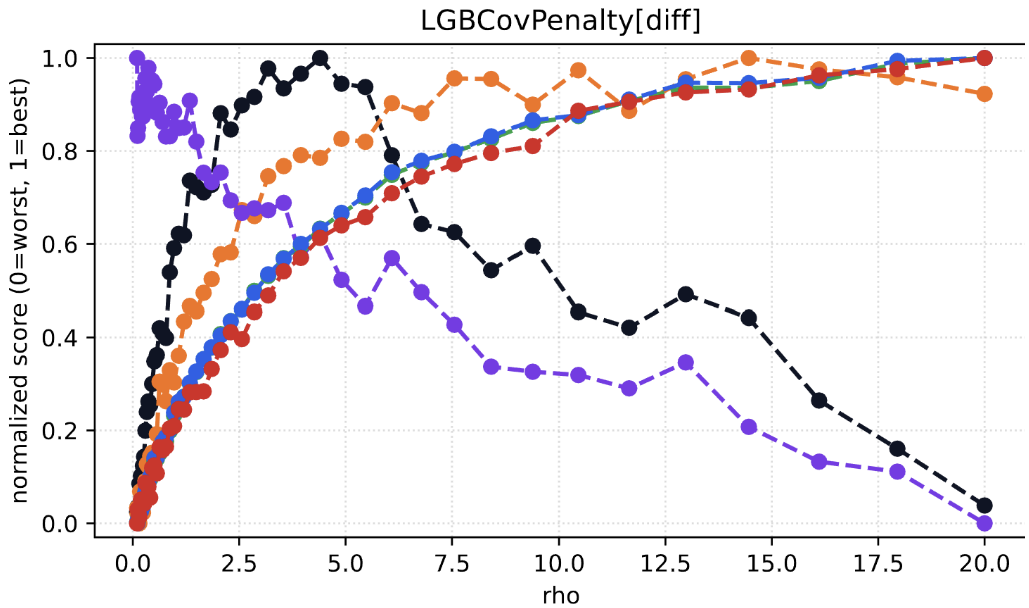
Validation Average

Metrics:

Data Year	**Baseline Model**	**Validation Avg Behavior**
2022	top1	

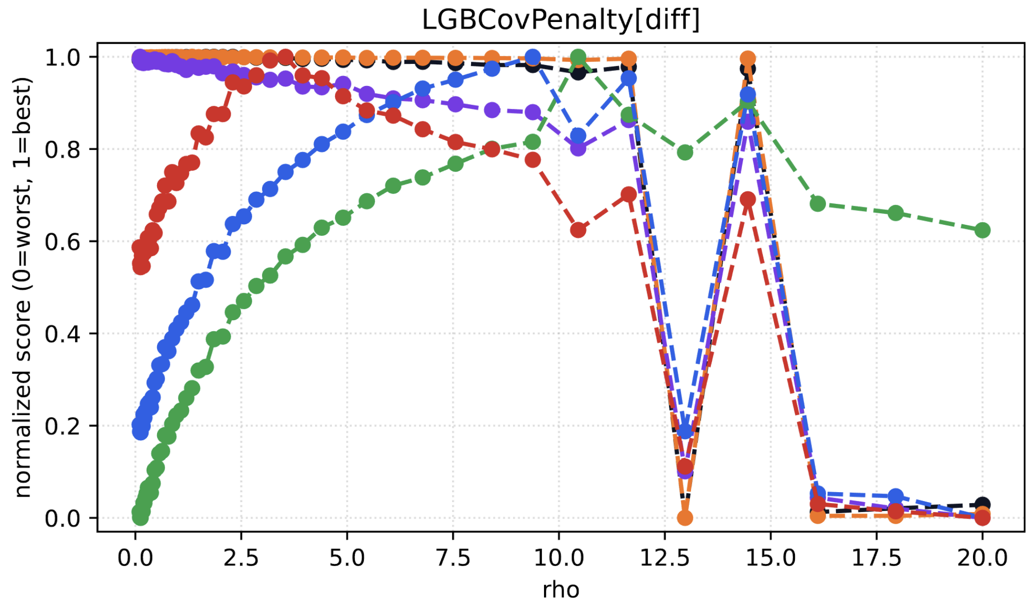
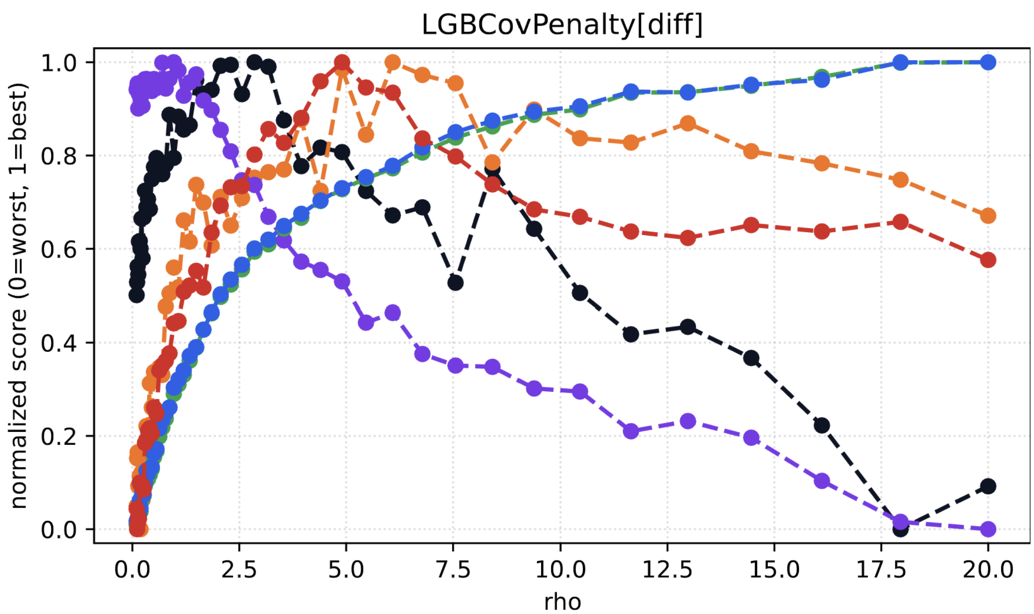
Data Year	Baseline Model	Validation Avg Behavior
2022	best r2	
2023	top1	

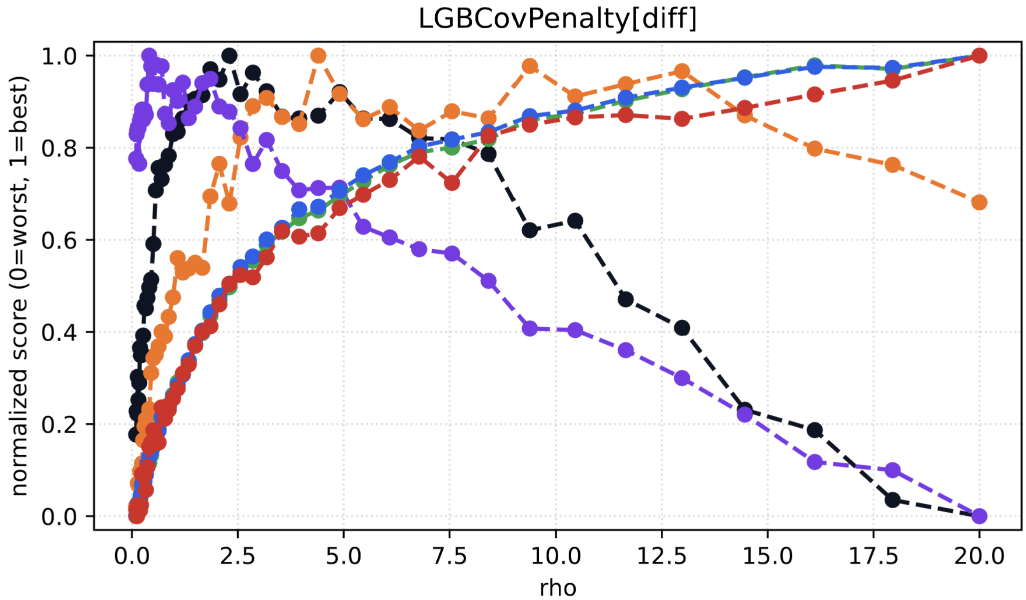
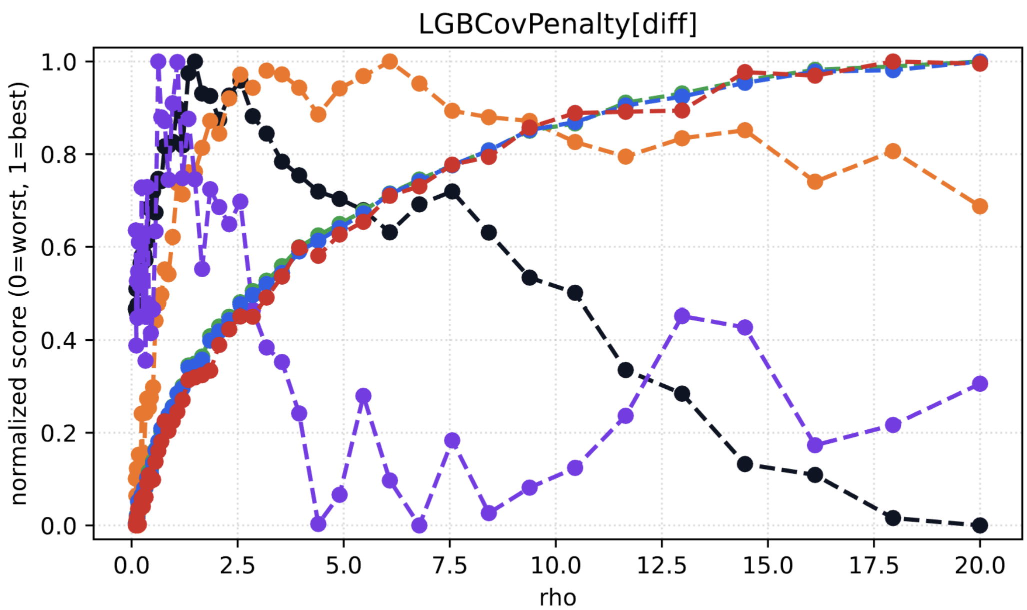
Data Year	Baseline Model	Validation Avg Behavior
2023	best r2	
2024 (old)	top1	
2024 (old)	best r2	

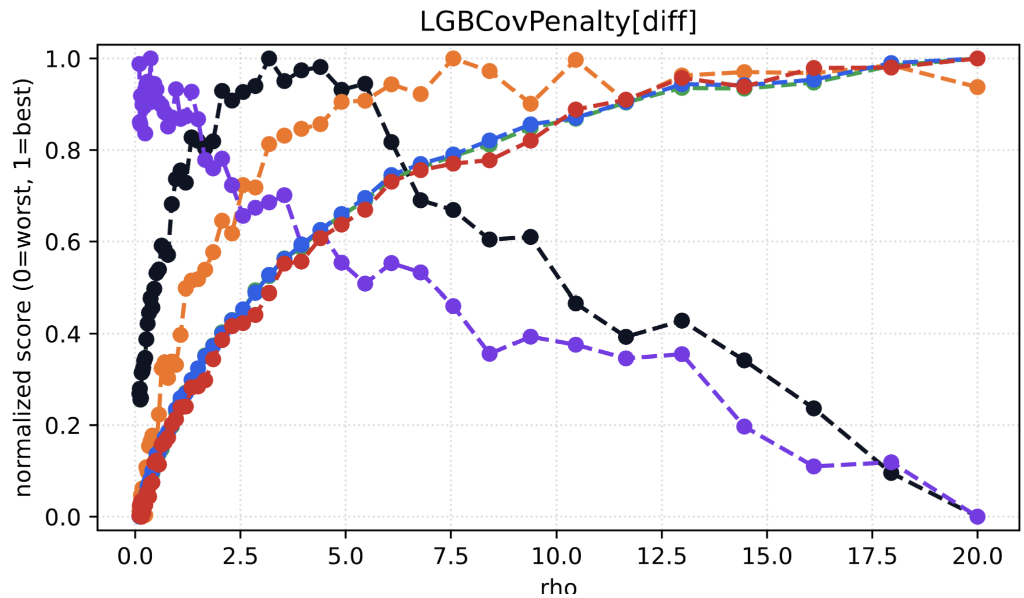
Data Year	Baseline Model	Validation Avg Behavior
2025	top1	
2025	best r2	

Test Average

Metrics:

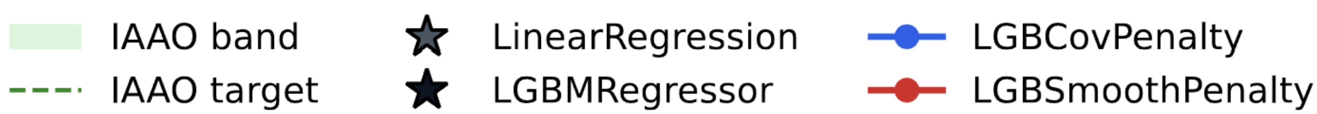
Data Year	Baseline Model	Test Behavior
2022	top1	
2022	best r2	

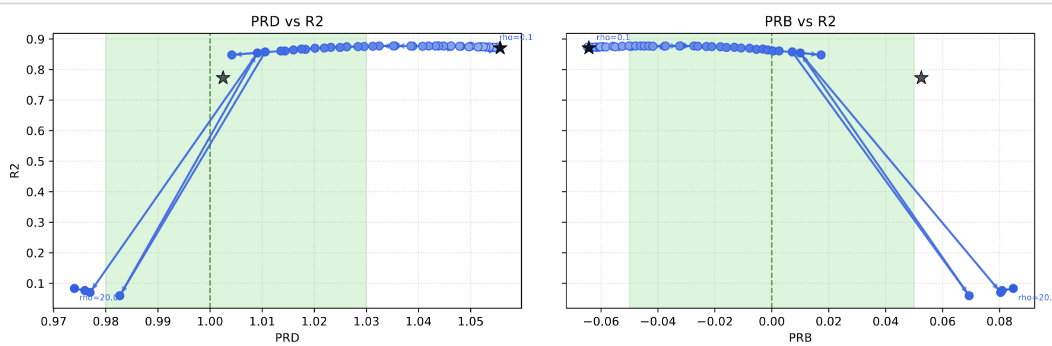
Data Year	Baseline Model	Test Behavior
2023	best r2	
2024 (old)	best r2	

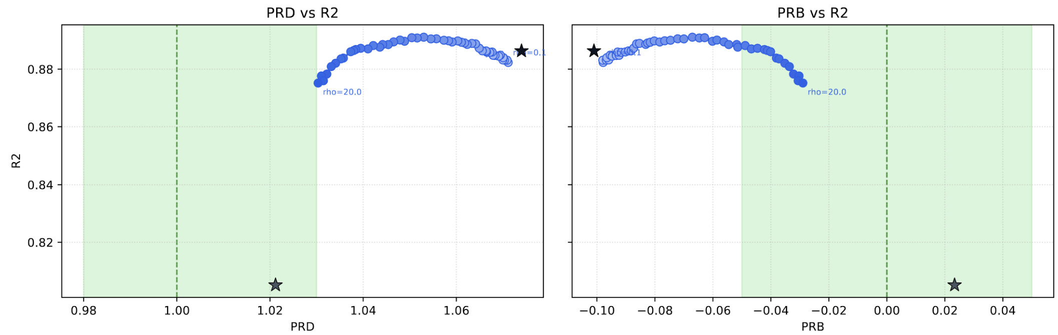
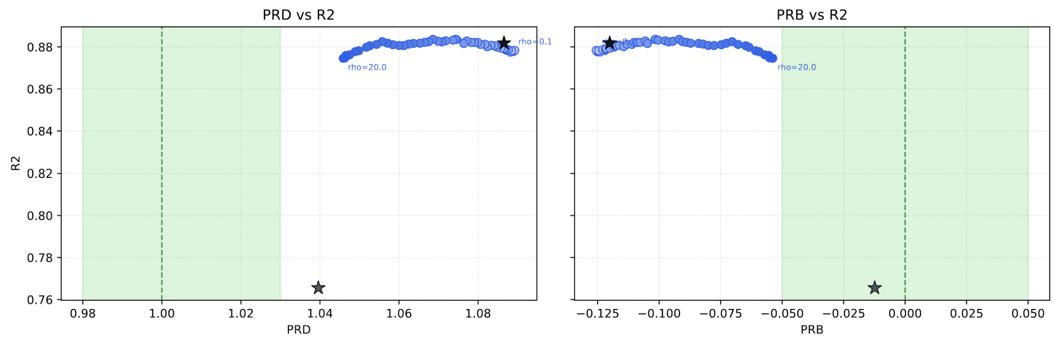
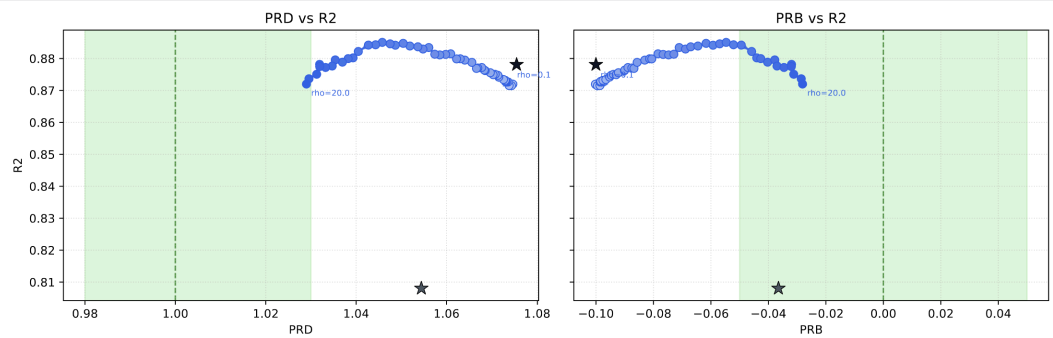
Data Year	Baseline Model	Test Behavior
2025	best r2	

2. Tradeoffs

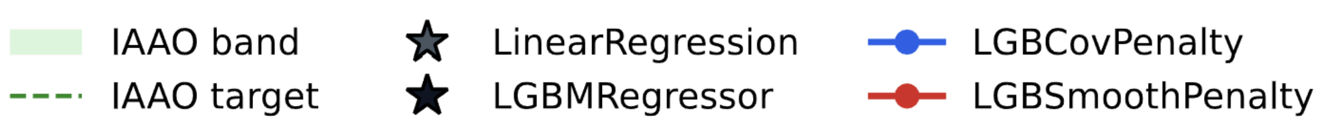
Validation Avg

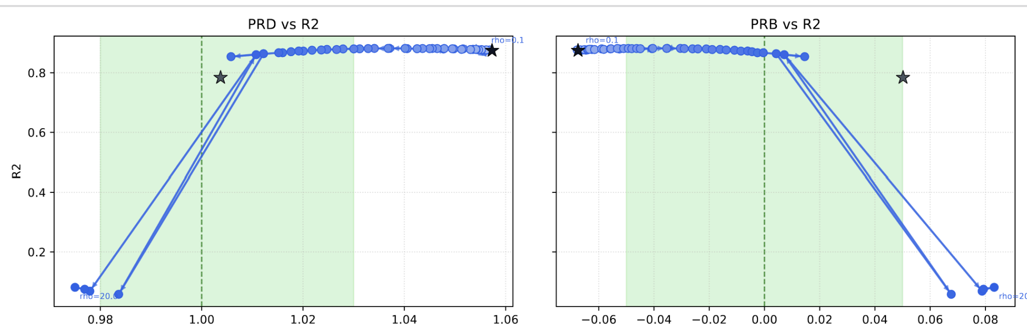


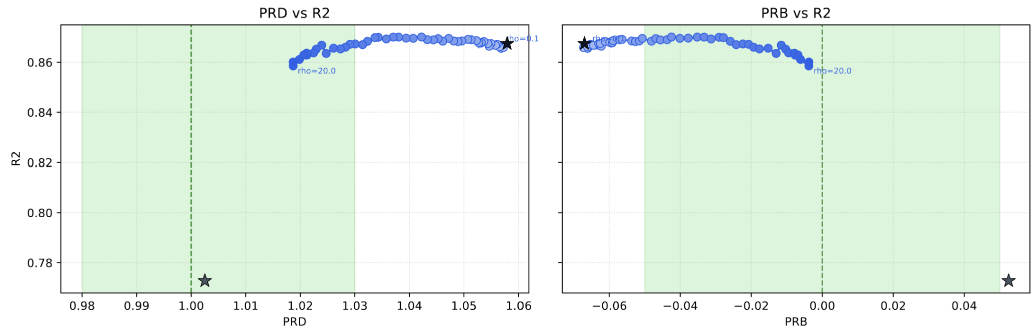
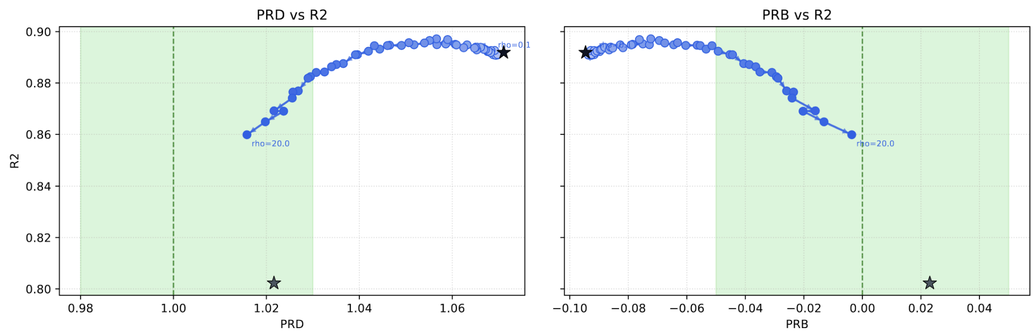
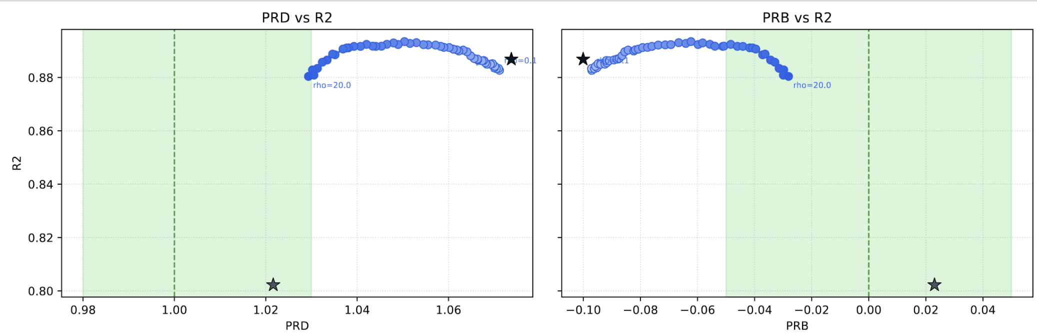
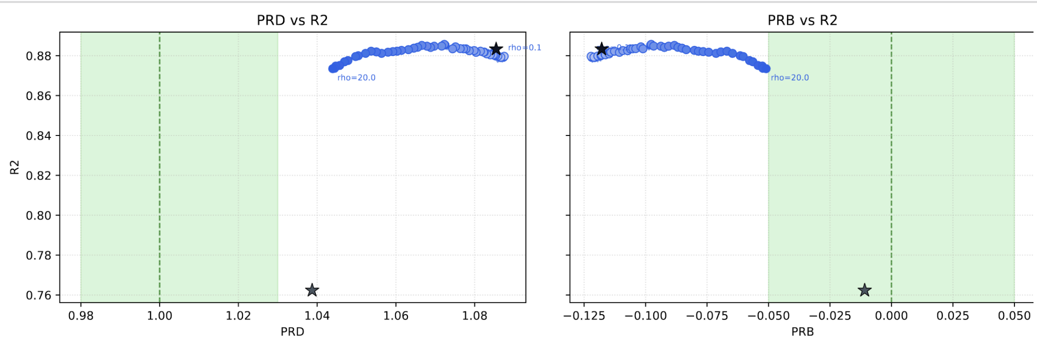
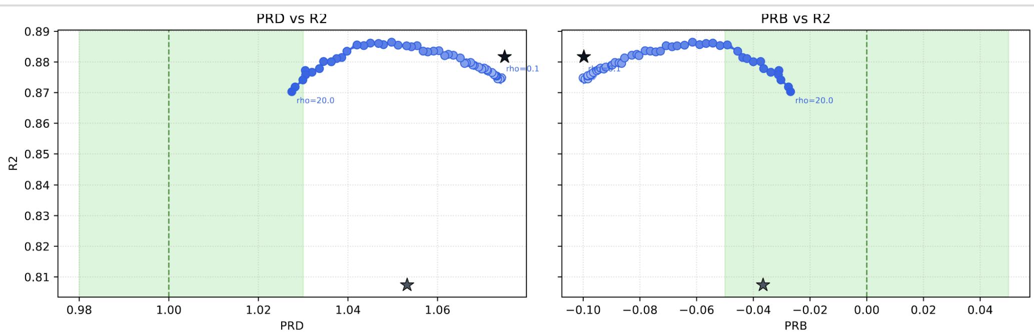
Data Year	Baseline Model	PRD/PRB vs MdAPE (Validation Avg)
2022	top1	

Data Year	Baseline Model	PRD/PRB vs MdAPE (Validation Avg)
2023	best r2	
2024	best r2	
2025	best r2	

Test



Data Year	Baseline Model	PRD/PRB vs MdAPE (Test)
2022	top1	

Data Year	Baseline Model	PRD/PRB vs MdAPE (Test)
2022	best r2	
2023	top1	
2023	best r2	
2024 (old)	best r2	
2025	best r2	

Notes

Tags?

#RA #MIT #RegressivePropertyTaxation